
Problem Set 3: Complex Numbers

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PY 355 Section A1

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Problem 1

1. Cartesian form:

$$z_1 = \frac{5}{3+4i} \cdot \frac{3-4i}{3-4i} = \frac{15+20i}{25} = \frac{3}{5} + \frac{4}{5}i.$$

Polar form:

$$|z_1| = \sqrt{\frac{9+16}{25}} = 1.$$

$$\theta = \arctan \frac{4}{3} \approx 0.927.$$

$$z_1 = e^{i \arctan 4/3} \approx e^{0.927i}.$$

2. Cartesian form:

$$z_2 = (3+4i)^2 = 9 - 16 + 24i = -7 + 24i.$$

Polar form:

$$|z_2| = \sqrt{49 + 576} = \sqrt{625} = 25.$$

$$\theta = \arctan -\frac{24}{7} \approx -1.287.$$

Arctan isn't defined on the proper region of $\mathbb{R}$ so we use the negative of this number.

$$z_2 \approx 25e^{1.287i}.$$

3. Cartesian form:

$$z_3 = \frac{1+\sqrt{3}i}{1-\sqrt{3}i} \cdot \frac{1+\sqrt{3}i}{1+\sqrt{3}i} = \frac{1-3+2\sqrt{3}i}{4} = -\frac{1}{2} + \frac{\sqrt{3}}{2}i.$$

Polar form:

$$|z_3| = \sqrt{\frac{1}{4} + \frac{3}{4}} = 1.$$

$$\theta = \arctan(\sqrt{3}) = \frac{2\pi}{3}.$$

$$z_3 = e^{i2\pi/3}.$$

4. Eulers formula. The number is already in cartesian form, and its polar form is $e^{i\theta}$.

5. Polar form:

$$\ln z_5 = i \ln 2 \implies z_5 = e^{i \ln 2}.$$

Cartesian form:

$$z_5 = \cos(\ln 2) + i \sin(\ln 2).$$

6. Cartesian form: note that

$$\frac{1+i}{1-i} = \frac{1-1+2i}{1+1} = i.$$

Thus,

$$\ln \frac{1+i}{1-i} = \ln i.$$
$$e^{i\theta} = i \implies \theta = \frac{\pi}{2}.$$

Therefore,

$$z_6 = i \frac{\pi}{2}.$$

Polar form:

$$i \frac{\pi}{2} = \frac{\pi}{2} \cos \frac{\pi}{2} + \frac{\pi}{2} i \sin \frac{\pi}{2} = \frac{\pi}{2} e^{i\pi/2}.$$

Problem 2

1.

$$z_1 + z_2 = \frac{3}{5} + \frac{4}{5}i - 7 + 24i = \frac{-32}{35} + \frac{124}{120}i.$$
$$z_1 z_2 \approx \exp(i(1.287 + 0.927)).$$
$$\frac{z_1}{z_2} \approx \exp(i(0.927 - 1.287)).$$

2.

$$z_6^2 = \frac{\pi^2}{4} i^2 = -\frac{\pi^2}{4}.$$

Problem 3

The book says

$$V(t) = L \frac{d^2 Q}{dt^2} + R \frac{dQ}{dt} + \frac{Q}{C}.$$

Given our definitions, and the fact that since Q is the charge, $I(t) = dQ/dt$, we have

$$I_0 Z(\omega) e^{i\omega t} = Li\omega I_0 e^{i\omega t} + RI_0 e^{i\omega t} + \frac{Q}{C}.$$

It follows that

$$Z(\omega) = Li\omega + R + \frac{Q}{C} I_0^{-1} e^{-i\omega t}.$$

Now note that

$$Q(t) = \int I(t) dt = \int I_0 e^{i\omega t} dt = \frac{I_0}{i\omega} e^{i\omega t}.$$

Plugging this in, we have

$$Z(\omega) = i\omega L + R + \frac{I_0}{I_0 C} \frac{e^{i\omega t}}{e^{i\omega t} i\omega} = i\omega L + R + \frac{1}{i\omega C}.$$

Problem 4

1. Let $z = x + iy \in \mathbb{C}$.

$$|z| = \sqrt{x^2 + y^2} \implies |z|^2 = x^2 + y^2 \geq x^2,$$

because $y \in \mathbb{R}$ implies $y^2 \geq 0$. Therefore, $|z| \geq x$. You can also note that

$$|z|^2 = x^2 + y^2 \geq y^2$$

by the same logic, so $|z| \geq y$. Note that $x = \operatorname{Re} z$ and $y = \operatorname{Im} z$.

2. I don't know if we're allowed to just say $|zw| = r_1 r_2 = |z| |w|$ with r_1, r_2 the magnitudes in polar coordinates so I did it with the cartesian representation. Let $z = a + bi$, $w = c + di$. It follows that

$$\begin{aligned} |zw| &= \sqrt{(ac - bd)^2 + (ad + bc)^2} \\ &= \sqrt{(ac)^2 - 2abcd + (bd)^2 + (ad)^2 + 2abcd + (bc)^2} \\ &= \sqrt{(ac)^2 + (ad)^2 + (bc)^2 + (bd)^2}. \end{aligned}$$

It follows that

$$\begin{aligned} |z| |w| &= \sqrt{a^2 + b^2} \sqrt{c^2 + d^2} \\ &= \sqrt{(a^2 + b^2)(c^2 + d^2)} \\ &= \sqrt{a^2 c^2 + a^2 d^2 + b^2 c^2 + b^2 d^2} \\ &= \sqrt{(ac)^2 + (ad)^2 + (bc)^2 + (bd)^2} \\ &= |zw|. \end{aligned}$$

$$|zw| = r_1 r_2 = |z| |w|.$$

3. Let $z = a + bi$, $w = c + di$. Note that

$$zw^* = (a + bi)(c - di) = (ac + bd) + (bc - ad)i.$$

Therefore, $\operatorname{Re} zw^* = ac + bd$. We have

$$|z + w|^2 = (a + c)^2 + (b + d)^2 = a^2 + b^2 + c^2 + d^2 + 2ac + 2bd = |z|^2 + |w|^2 + 2 \operatorname{Re} zw^*.$$

Problem 5

$$e^{i(A+B)} = e^{iA} e^{iB}$$

$$\begin{aligned} \implies \cos(A+B) + i \sin(A+B) &= (\cos A + i \sin A)(\cos B + i \sin B) \\ &= \cos A \cos B - \sin A \sin B + (\cos A \sin B + \cos B \sin A) i. \end{aligned}$$

Therefore,

$$\cos(A+B) = \cos A \cos B - \sin A \sin B,$$

$$\sin(A+B) = \cos A \sin B + \cos B \sin A.$$